The Project Management Process
Lecture 6

Project Management Process Groups

A **process** is a series of actions directed toward a particular result
Project management can be viewed as a number of interlinked processes
The project management process groups include:

- Initiating processes
- Planning processes
- Executing processes
- Monitoring and controlling processes
- Closing processes

Project Management Process Groups and Knowledge Area Mapping*

Knowledge Area	Project Management Process Groups				
	Initiating	Planning	Executing	Monitoring and Controlling	Closing
Project Integration Management	Develop project charter	Develop project management plan	Direct and manage project execution	Monitor and control project work, Perform integrated change control	Close project or phase
Project Scope Management		Collect requirements, Define scope, Create WBS		Verify scope, Control scope	
Project Time Management		Define activities, Sequence activities,		Control schedule	

*Source: PMBOK® Guide, Fourth Edition, 2008.

Project Management Process Groups and Knowledge Area Mapping*

Knowledge Area	Project Management Process Groups				
	Initiating	Planning	Executing	Monitoring and Controlling	Closing
Project Time Management (continued)		Estimate activity resources, Estimate activity durations, Develop schedule			
Project Cost Management		Estimate costs, Determine budget		Control costs	
Project Quality Management		Plan quality	Perform quality assurance	Perform quality control	
Project Human Resource Management		Develop human resource plan	Acquire project team, Develop project team, Manage project team		

Project Management Process Groups and Knowledge Area Mapping

Knowledge Area	Project Management Process Groups				
	Initiating	Planning	Executing	Monitoring and Controlling	Closing
Project Communications Management	Identify stakeholders	Plan communications	Distribute information, Manage stakeholders expectations	Report performance	
Project Risk Management		Plan risk management, Identify risks, Perform qualitative risk analysis, Perform quantitative risk analysis, Plan risk responses		Monitor and control risks	
Project Procurement Management		Plan procurements	Conduct procurements	Administer procurements	Close procurements

Project Integration Management

- Processes and activities needed to identify, define, combine, unify, and coordinate the various processes and project management activities within the project management process groups.
- It **does not** refer to the technical process of integrating software components to form a partial or completed software product.
- Software project managers are responsible for planning and conducting a wide scope of project activities.

Project Integration Management Processes

- Process 1: Develop Project Charter
- Process 2: Develop Project Management Plan
- Process 3: Direct and Manage Project Work
- Process 4: Monitor and Control Project Work
- Process 5: Perform Integrated Change Control
- Process 6: Close Project or Phase

Process 1: Develop Project Charter

- Developing a document that formally authorizes the existence of a project.
- Key benefits
  - Well-defined project start and project boundaries
  - Creation of a formal record of the project
  - Direct way for senior management to formally accept and commit to the project



Develop Project Charter: Inputs [1/2]

- Project Statement of Work (SOW)
  - Narrative description of products, services, or results
  - Provided by project initiator or sponsor (internal and external)
  - Business need
    - Market demand
    - Technology advancement
    - Legal requirement
    - Govt. regulation
    - Environmental consideration
  - Product scope description
  - Strategic plan
- Business case (Justification of business need)
  - Business viewpoint of the project
  - Business need and cost-benefit analysis
  - Total cost of ownership (operational and sustainment costs)

Develop Project Charter: Inputs [2/2]

- Agreements
  - Initial intentions for a project
  - Contracts, MoUs etc.
- Enterprise environmental factors
  - Govt./industry standards, marketplace conditions etc.
- Organizational process assets
  - Organizational standard processes, policies, and process definitions
  - Templates (e.g., project charter template)
  - Historical information and lessons learned knowledge base

Develop Project Charter: Tools and Techniques

- Expert judgment
  - Applied to technical and management details
  - Provided by any group or individual with special knowledge/training
  - Other units within the organization, consultants, and stakeholders etc.
  - Domain experts
- Facilitation techniques
  - Brainstorming
  - Conflict resolution
  - Problem solving
  - Meeting management

Develop Project Charter: Outputs

- Project charter
  - **Document** issued by the project sponsor or initiator
  - Project purpose or justification
  - Measurable project objectives and related success criteria
  - High-level requirements
  - High-level project description and boundaries
  - High-level risks
  - Summary milestone schedule
  - Summary budget
  - Stakeholder list
  - Project approval requirements
  - Assigned project manager, responsibility, and authority level
  - Name and authority of the sponsor or other person(s) authorizing the project charter
- Assumption Log
  - Both Assumptions and Constraints

Process 2: Develop Project Management Plan

- The process of defining,preparing, and coordinating all subsidiary plans and integrating them into a comprehensive project management plan
- How the project is executed, monitored and controlled, and closed



Develop Project Management Plan: Inputs

- Project charter
- Outputs from other processes
- Enterprise environmental factors
- Organizational process assets
  - Standardized guidelines, work instructions, proposal evaluation criteria, and performance measurement criteria
  - Project management plan template
  - Change control procedures
  - Project files from previous projects
  - Historical information and lessons learned knowledge base
  - Configuration management knowledge base containing the versions and baselines of all official organization standards, policies, procedures, and any project documents.

Develop Project Management Plan: Tools and Techniques

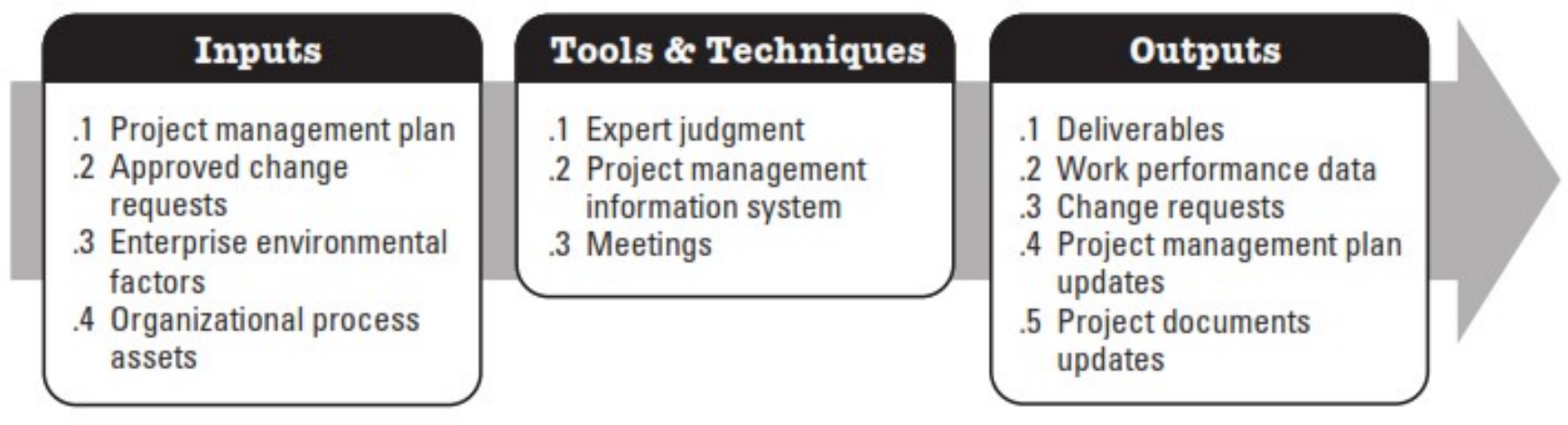
- Expert judgment
  - Tailor the process to meet the project needs
  - Develop technical and management details to be included in the project management plan
  - Determine resources and skill levels needed to perform project work
  - Define the level of configuration management to apply on the project
  - Determine which project documents will be subject to the formal change control process
  - Prioritize the work on the project to ensure the project resources are allocated to the appropriate work at the appropriate time.
- Facilitation techniques

Develop Project Management Plan: Outputs

- Project management plan
  - Project baselines
  - Requirements management plan
  - Configuration management plan
  - Security plans (physical, project, data)
  - Quality management plan
  - Enterprise technology insertion plan
  - Information security plan
  - Test and evaluation plan
  - Information management plan
  - Release and deployment plan
  - Technology infrastructure plan
  - Project-specific training plan for the software team

Process 3: Direct and Manage Project Work

- Leading and performing the work defined and implementing approved changes



4.3 Direct and Manage Project Work

.1 Inputs
- .1 Project management plan
- .2 Approved change requests
- .3 Enterprise environmental factors
- .4 Organizational process assets

.2 Tools & Techniques
- .1 Expert judgment
- .2 Project management information system
- .3 Meetings
- ***.4 Information dissemination***

.3 Outputs
- .1 Deliverables
- *.2 Work performance data*
- .3 Change requests
- .4 Project management plan updates
- .5 Project documents updates
- ***.6 Demonstrations of working, deliverable software***

Figures sources: PMBOK® Guide - Fifth Edition, PMI, 2013, pp. 79 & Software Extension to PMBOK® Guide, PMI, 2013, pp. 46

Direct and Manage Project Work: Activities (adaptive life cycle) [1/2]

- **Communicating** the project scope, resources, and schedule/budget constraints to the project team and other appropriate stakeholders;
- Providing the **resources** and facilities needed by the **software team**
- Managing **project scope**, **stakeholder expectation**, **resources**, **schedule**, and **budget**
- Creating effe**ctive mechanisms** to see an ongoing overall picture of progress, status, and issues
- **Controlling** changes to project scope, resources, schedule, and budget
- **Consulting** with and allocating the work among multiple software development teams
- Ensuring communication and coordination of work activities among multiple teams and coordinating the integration of resources and components
- Facilitating ongoing and continuous communications between the customer/user representatives and the software developers
- Monitoring productivity, product quality, and team performance while making adjustments as needed

Direct and Manage Project Work: Activities (adaptive life cycle) [2/2]

- Ensuring **effective interactions** with other organizational units, such as independent testing and user training
- **Facilitating demonstrations** of evolving product capabilities for appropriate stakeholders
- Facilitating **delivery of early product** capabilities into the user environment, as desired
- Facilitating delivery and **acceptance of the final product** and closing the project
- Coordinating dependencies with related projects, which may be elements of a common development program or milestone dependencies with related IT infrastructure projects
- **Managing risk**

Direct and Manage Project Work: Inputs

- Project management plan
- Approved change requests
- Enterprise environmental factors
  - Organizational, company, or customer culture and structure of the performing or sponsor organizations
  - Infrastructure (e.g., existing facilities and capital equipment)
  - Personnel administration (e.g., hiring and firing guidelines, and employee performance reviews)
  - Stakeholder risk tolerances, for example allowable cost overrun percentage
  - Project management information system
- Organizational process assets

Direct and Manage Project Work: Tools and Techniques

- Expert judgment
- Project management information system
- Meetings
- Information dissemination
  - Current overall status of the project
  - Risks and risk status (watch list, monitored, confronted)
  - Current work assignments
  - Daily progress and remaining work
  - Forecasts of future project status
  - Number of requirements, features, stories, or use cases written/demonstrated/delivered
  - Number of **test cases and test scenarios** written/passed
  - Product components/features implemented versus cost or staff-hours
  - Resolutions and action items from the last retrospective meeting
  - Status of servers and other infrastructure equipment (up, down, in maintenance)

Direct and Manage Project Work: Outputs

- Deliverables
- Work performance data
- Change requests
  - Corrective action
  - Preventive action
  - Defect repair
  - Updates
- Project management plan updates
- Project documents updates
- Demonstration of working, deliverable software


ANY
QUESTIONS